

## Response Letter with Annotations

Chen, A., Chen, K., Zhang, J., Meng, J., & Shen, C. (2023). When national identity meets conspiracies: The contagion of national identity language in public engagement and discourse about COVID-19 conspiracy theories. *Journal of Computer-Mediated Communication*, 28(1), zmac034. <https://doi.org/10.1093/jcmc/zmac034>

### Annotations

#### 1. Purposes

##### a. Motivation (one-sentence summary)

- This study is motivated by the question of how national identity language in conspiracy-related posts influences engagement and spreads through online discussions and users' everyday communication.

##### b. Specific Rationale (several bullet points)

- Existing research have explored how message characteristics (e.g., emotions, animosity) can affect the virality of conspiracy theory on social media, little is known about how identity language influences engagement with conspiracies on social media.
- **SIDE** (Social identity model of deindividuation effects) **Model** provides a valuable framework for investigating how national identity narratives can influence public engagement with conspiracy theories and counternarratives.
  1. Previous work has explained the formation of **in-group identification and outgroup derogation** as forces that can drive individuals to affirm and defend their views.
  2. **Polarization** is strongest in computer-assisted group discussions when individuals are isolated in different rooms (anonymity) and are referred to using a group code (group salience).

##### c. Research Questions and/or Hypotheses

- Identify IV, DV, mediator, moderator
  1. IVs
    - a. National identity language in conspiracy-related posts:
      - i. In-group favoritism language
      - ii. Out-group hostility language
  2. DVs
    - a. User engagement size, including
      - i. Number of reposts
      - ii. Number of likes

- iii. Number of comments
    - iv. Diffusion size of conspiracy-related posts
  - b. Discourse contagion
    - i. Frequency of national identity language appearing in replies and comments following a conspiracy-related post
  - c. User-level communication traits
    - i. Average frequency of in-group favoritism and out-group hostility language in users' daily, non-conspiracy-related posts
- Identify specific relationships among variables
  1. H1: Conspiracy-related posts that use in-group favoritism language are likely to receive more likes, comments, and a larger diffusion size than those that do not use in-group favoritism language.
  2. H2: Conspiracy-related posts that use in-group favoritism language are likely to receive replies containing more ingroup favoritism language compared with posts that do not.
  3. H3(a): Conspiracy-related posts that use out-group hostility language are likely to receive fewer likes than those that do not. H3(b): Conspiracy-related posts that use out-group hostility language are likely to receive more comments and have larger diffusion compared with those that do not.
  4. H4: Conspiracy-theory-related posts that use out-group hostility language are likely to receive replies containing more out-group hostility language than posts that do not use out-group hostility language.
  5. H5: People who use national identity language (i.e., ingroup favoritism and out-group hostility words) when discussing COVID-19 conspiracy theories are also more likely to use such language in their everyday posts compared with those who do not use national identity language when discussing COVID-19 conspiracy theories.

## 2. Method

### a. Research Design

- Computational content analysis: data source, etc.
  1. Data source:
    - a. Platform: Weibo
    - b. Dataset: COVID-19-related public posts

### b. Sample

- Data collection
  1. First, they retrieved more than 40 million COVID-19-related posts. Second, they drew from mainstream factchecking websites to identify 35 types of COVID-19 conspiracy-related narratives

which were used as the basis for classifying whether a post qualified as containing a conspiracy. Third, they had three coders to develop keyword combinations to summarize each conspiracy narrative to retrieve posts using a Boolean query search method.

2. Three trained coders developed keyword combinations summarizing each conspiracy narrative. These keyword combinations were used to retrieve posts through a Boolean query search.
3. After the above three steps, these COVID-19 conspiracy theories and their representative keyword combinations allowed authors to retrieve 6,735 originally crafted conspiracy-related posts (which sparked a total of 346,783 reposts and 38,075 replies) from 6,735 unique users.

c. Measures

- User engagement size
  1. including the number of reposts, likes, and comments, as well as the extent of diffusion of a conspiracy theory post.
- Discourse
  1. discussions around a conspiracy theory post (in the form of comments and replies to the original post/reposts), which forms a conversation between users with similar interests in the conspiracy narratives.
- The traits of users
  1. average frequency with which these users utilize in-group favoritism words and out-group hostility words in their daily posts.
    - a. Average frequency of in-group favoritism words
    - b. Average frequency of out-group hostility words
  2. Calculated from user's daily posts outside the conspiracy context

d. Analysis

- Dictionary-based text analysis used to identify national identity language
- Poisson Regression models
- T-tests

### 3. Results and Conclusions

a. Answers to RQs and Hypotheses

- Conspiracy-related posts containing stronger national identity language generate **higher engagement and wider diffusion**.
- National identity language **spreads through discourse**, appearing in replies and comments following conspiracy-related posts.
- Users who participate in conspiracy discussions are more likely to **use national identity language in their broader, everyday communication**.

b. Include Key Figures/Tables

**Table 1.** Overview of the national identity dictionary

| Dimensions                | Examples                                                                                                                                                                                                                                  |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| In-group favoritism words | 炎黄子孙 (descendants of Civilized China), 龙的传人 (descendants of the dragon), 中华 (greater China), 华夏 (civilized China), 先辈 (ancestors), 先烈 (martyrs), 祖国 (motherland), 中国梦 (Chinese dream), and 长江文明 (Yangtze River civilization)              |
| Out-group hostility words | 帝国 (empire), 洋人 (outlanders), 外国势力 (foreign powers), 国外势力 (foreign powers), 西方势力 (Western powers), 境外势力 (foreign forces), 白左 (white left), 八国联军 (the Eight-Power Allied Forces), 列强 (imperialist powers), 昂撒 (Angsa), and 白皮猪 (white pig) |

**Table 2.** Models selected to examine hypotheses

| Hypotheses                                                | Independent variables                                                                                 | Dependent variables                                                                          | Model selected     |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------|
| H1, H3a, and H3b<br>(results in <a href="#">Table 3</a> ) | a) In-group favoritism and<br>b) out-group hostility language in conspiracy-theory-related posts      | User engagement size (i.e., number of likes, comments, and diffusion size)                   | Poisson regression |
| H2 ( <a href="#">Table 4</a> )                            | a) In-group favoritism or<br>b) out-group hostility language in conspiracy-theory-related posts       | In-group favoritism word count in replies                                                    | Poisson regression |
| H4 ( <a href="#">Table 4</a> )                            | a) In-group favoritism or<br>b) out-group hostility language in conspiracy-theory-related posts       | Out-group hostility word count in replies                                                    | Poisson regression |
| H5                                                        | a) In-group favoritism or<br>b) out-group hostility language adoption (i.e., adopter vs. non-adopter) | (a) In-group favoritism or<br>(b) out-group hostility average word count in historical posts | <i>t</i> -tests    |

Notes: (1) Adopters are those who used (a) in-group favoritism or (b) out-group hostility language in their discussions about conspiracy-related posts, while non-adopters are those who did not use such national identity language. (2) The rationale for model selection is explained in [Supplementary Appendix G](#).

**Table 3.** Poisson regression models predicting user engagement size of conspiracy-related posts on Weibo

|                                                              | Model 1a        |       | Model 1b             |       | Model 1c       |       |
|--------------------------------------------------------------|-----------------|-------|----------------------|-------|----------------|-------|
|                                                              |                 |       | User engagement size |       |                |       |
|                                                              | Number of likes |       | Number of comments   |       | Diffusion size |       |
|                                                              | <i>b</i>        | SE    | <i>b</i>             | SE    | <i>b</i>       | SE    |
| Use of national identity language                            |                 |       |                      |       |                |       |
| In-group favoritism (ref. without in-group favoritism words) | 0.151***        | 0.020 | 0.074***             | 0.035 | 0.230***       | 0.045 |
| Out-group hostility (ref. without out-group hostility words) | −0.328***       | 0.033 | 0.116***             | 0.049 | 0.160*         | 0.071 |
| Control variables                                            |                 |       |                      |       |                |       |
| Content-level controls                                       |                 |       |                      |       |                |       |
| Debunking post (ref. conspiracy post)                        | 0.555***        | 0.018 | 0.633***             | 0.032 | 0.739***       | 0.041 |
| Text length                                                  | 0.002***        | 0.001 | 0.002***             | 0.001 | 0.002***       | 0.001 |
| URL (ref. no URL)                                            | 0.135**         | 0.018 | 0.210***             | 0.031 | 0.509***       | 0.038 |
| Hashtag count                                                | 0.134***        | 0.004 | 0.113***             | 0.006 | 0.145***       | 0.008 |
| Text sentiment score                                         | −0.022***       | 0.001 | −0.014***            | 0.001 | −0.017***      | 0.002 |
| User-level controls                                          |                 |       |                      |       |                |       |
| Verified individual users (ref. = ordinary user)             | −0.004          | 0.023 | −0.1017***           | 0.035 | −0.771***      | 0.036 |
| Verified organizational users (ref. = ordinary user)         | −0.620***       | 0.028 | −1.748***            | 0.045 | −1.611***      | 0.050 |
| Gender of user (ref. = female)                               | −0.201***       | 0.018 | −0.604***            | 0.031 | −0.607***      | 0.039 |
| User's follower number                                       | 0.537***        | 0.003 | 0.591***             | 0.005 | 0.744***       | 0.007 |
| User's followee number                                       | −0.054          | 0.008 | −0.080***            | 0.015 | −0.175***      | 0.018 |
| User's number of historical posts                            | −0.271***       | 0.007 | −0.179***            | 0.011 | −0.319***      | 0.014 |
| Constant                                                     | −0.173**        | 0.053 | −1.758***            | 0.093 | −2.706***      | 0.126 |
| Pseudo- $R^2$                                                | 49.7%           |       | 54.8%                |       | 74.0%          |       |
| N                                                            |                 |       | 1,516                |       |                |       |

Notes: (1) Reference group: gender = female, verified = unverified Weibo user; (2) Activeness, # of followers, and # of followees have been log-transformed; (3) \* $p < .05$ , \*\* $p < .01$ , \*\*\* $p < .001$ ; (4) SE: standard error; (5) The beta coefficients can be interpreted as the log of the ratio of expected counts.

**Table 4.** Poisson regression models predicting the use of national identity language in replies

|                                                                       | Model 2a                       |       | Model 2b                                               |       |
|-----------------------------------------------------------------------|--------------------------------|-------|--------------------------------------------------------|-------|
|                                                                       |                                |       | National identity language usage in subsequent replies |       |
|                                                                       | In-group favoritism in replies |       | Out-group hostility in replies                         |       |
|                                                                       | <i>b</i>                       | SE    | <i>b</i>                                               | SE    |
| Use of national identity language                                     |                                |       |                                                        |       |
| In-group favoritism in posts (ref. without in-group favoritism words) | 1.823***                       | 0.027 | −0.391                                                 | 0.030 |
| Out-group hostility in posts (ref. without out-group hostility words) | −0.201***                      | 0.027 | 1.701***                                               | 0.027 |
| Control variables                                                     |                                |       |                                                        |       |
| Content-level controls                                                |                                |       |                                                        |       |
| Debunking post (ref. conspiracy post)                                 | 0.094***                       | 0.048 | 0.326***                                               | 0.048 |
| Text length                                                           | 0.024***                       | 0.001 | 0.021***                                               | 0.001 |
| URL (ref. no URL)                                                     | −0.790*                        | 0.197 | 0.272*                                                 | 0.138 |
| Hashtag count                                                         | −0.014*                        | 0.006 | 0.050                                                  | 0.006 |
| Text sentiment score                                                  | −0.016***                      | 0.002 | −0.012***                                              | 0.002 |
| User-level controls                                                   |                                |       |                                                        |       |
| Verified individual users (ref. = ordinary user)                      | −0.170***                      | 0.034 | 0.007                                                  | 0.035 |
| Verified organizational users (ref. = ordinary user)                  | 0.030                          | 0.043 | −0.086*                                                | 0.044 |
| Gender of user (ref. = female)                                        | 0.314***                       | 0.034 | −0.376***                                              | 0.030 |
| User's follower number                                                | 0.010*                         | 0.005 | 0.057***                                               | 0.005 |
| User's followee number                                                | −0.021                         | 0.010 | 0.006                                                  | 0.010 |
| User's number of historical posts                                     | 0.031***                       | 0.008 | 0.039***                                               | 0.008 |
| Constant                                                              | −2.753***                      | 0.084 | −3.020***                                              | 0.086 |
| Pseudo- $R^2$                                                         | 52.9%                          |       | 53.0%                                                  |       |
| N                                                                     |                                |       | 17,405                                                 |       |

### c. Authors' Conclusions

- National identity language plays a key role in amplifying the reach and impact of conspiracy theories.
- Conspiracy discussions function as a mechanism through which identity-based language becomes contagious and extends beyond isolated events into users' broader communication behavior.

## 4. Discussion

### a. Implications

- Demonstrates that **conspiracy theories are not only informational but also identity-laden**, with national identity language playing a central role in their diffusion.
- Shows that identity language amplifies engagement and discourse contagion, helping conspiracy narratives spread beyond isolated posts.
- Suggests that conspiracy participation can spill over into users' broader communication styles, reinforcing identity-based discourse over time.

### b. Limitations

- The study focuses on **text-based content** and does not capture visual or multimodal conspiracy content.
- National identity is measured through **dictionary-based word counts**, which may not fully capture contextual uses of identity language.
- The observational design limits **causal inference**.
- Findings are based on **Weibo and the COVID-19 context**, which may limit generalizability to other platforms or issues.

### c. Future Directions

- It would be valuable for future research to identify institutional-hired nationalists versus individual based nationalists to examine the impact of their posts separately because both actors shape discourses on Chinese social media.
- Examine visual and multimodal forms of conspiracy content and their identity cues.
- Explore causal mechanisms through experiments or mixed-method approaches.
- Investigate how other forms of identity (e.g., political, religious, ethnic) interact with conspiracy narratives.
- Study the long-term effects of identity-based discourse on users' political attitudes and behaviors.

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**Response Letter (~600 words)**

- Your assessment of the study
  - One major strength of this study lies in the unusually tight connection between theory and data. Rather than invoking social identity concepts in a broad or generic way, the authors ground their analysis explicitly in the social identity model of collective action, which allows them to move beyond a simple in-group versus out-group distinction. For me, this was especially striking, because discussions of in-group and out-group dynamics are often limited to general expectations about favoritism and hostility. In contrast, this study operationalizes these theoretical ideas through carefully constructed measures of identity language and uses them to examine concrete behavioral and discursive outcomes. Even though the paper does not explicitly position itself as filling all theoretical gaps, the choice of theory is highly appropriate for the research.
  - A second major strength is the rigor of the methodological approach. The authors do not rely on a single computational dictionary but instead combine multiple strategies to ensure the validity of their measures. They construct a customized dictionary for national identity language, draw on mainstream fact-checking websites to systematically identify COVID-19 conspiracy narratives, and incorporate three human coders to classify conspiracy-related content, reporting high intercoder reliability. This hybrid approach, integrating computational text analysis with manual validation, demonstrates careful attention to measurement challenges in large-scale social media research.
- Weaknesses
  - One potential limitation concerns the conceptual treatment of conspiracy theories as misinformation. Conspiracy narratives often involve strong subjective intent and purposeful framing, which may align more closely with disinformation than with unintentional misinformation. By grouping conspiracy theories under the broader category of misinformation, the study does not differentiate between false information that spreads unintentionally and content that may be strategically produced or disseminated with specific goals. This conceptual ambiguity may limit how precisely the findings speak to different types of problematic information.
  - A second limitation concerns the potential influence of automated or coordinated accounts on the observed patterns of engagement and diffusion. This issue is particularly salient in the context of Weibo, where prior research and public reporting suggest the widespread presence of bots and organized online actors. Because the study relies on observational platform data, it is difficult to determine whether reposts, comments, and diffusion chains reflect organic human interaction or are partially driven by automated or strategically coordinated behavior. Without explicit procedures to detect or filter bot activity, engagement size and discourse contagion may be inflated by non-human or semi-automated accounts. This is especially relevant given that the study focuses on the diffusion of identity-laden conspiracy narratives, which may be disproportionately

amplified by coordinated actors seeking to shape public discourse. As a result, some of the observed contagion effects may reflect platform-specific dynamics.

- Other notes